

Paper-Faithful Imbalance Mitigation for TCGA-UT Frozen WSI Features

Abstract

Class imbalance is a persistent obstacle in computational pathology because clinically relevant cohorts often follow long-tailed disease distributions. This study evaluates imbalance-mitigation methods on TCGA-UT using a shared frozen Virchow2 whole-slide-image feature representation and identical stratified splits. The revised experiment is grounded in a five-part taxonomy: data-level augmentation, synthetic generation, algorithm-level losses, hybrid methods, and representation or architecture methods. Unlike a proxy-only comparison, each empirical method is implemented as a representative method that can operate on frozen WSI feature bags. Across three seeds, the strongest test macro F1 is achieved by focal loss (0.849), unweighted cross-entropy (0.846), balanced sampling (0.845), and weighted cross-entropy (0.844). The best representative bag method, RankMix-style MIL, reaches 0.825 macro F1, while the encoded image-GAN synthetic condition reaches 0.822. These results indicate that frozen-feature TCGA-UT performance is dominated by the strength of the Virchow2 representation and simple downstream losses, while more structured bag-level representatives remain sensitive to adaptation and resource constraints.

1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate the training set, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation reports only micro-averaged or prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings. [1–3]

This study treats TCGA-UT as a long-tailed computational pathology benchmark. The experiments preserve the observed class distribution in the training split and evaluate on validation and test splits drawn from the same distribution. This design asks which downstream mitigation methods help when class frequency is part of the cohort structure that the model must handle.

The revised comparison addresses a limitation of the earlier proxy sweep. Methods such as oversampling, feature interpolation, and center regularization are useful baselines, but they do not reproduce the methodological commitments of the representative papers in the taxonomy. The new experiment therefore evaluates WSI feature-bag implementations where possible and explicitly marks image-level or multimodal components that cannot be reproduced from the available frozen features.

2 Dataset

The dataset used in this experiment contains 8736 whole-slide images from TCGA-UT, assigned to 32 cancer-type labels after matching slide identifiers to available frozen Virchow2 feature tensors. The largest class is Glioblastoma multiforme with 792 slides, followed by Brain Lower Grade Glioma with 784 and Breast invasive carcinoma with 770. The smallest class is Lymphoid

Neoplasm Diffuse Large B-cell Lymphoma with 28 slides, followed by Cholangiocarcinoma with 30 and Pheochromocytoma and Paraganglioma with 45. The resulting slide-level imbalance ratio is 28.3:1. Stratified splits contain 6116 training slides, 1310 validation slides, and 1310 test slides.

This distribution is a useful example of the long-tailed class-imbalance problem because the skew is not limited to a single rare outlier. The eight largest classes account for 4565 slides, or 52.3% of the cohort, while the eight smallest classes account for only 444 slides, or 5.1%. Nine classes have fewer than 100 slides each. These classes are not interchangeable technical labels; they correspond to distinct diagnostic categories.

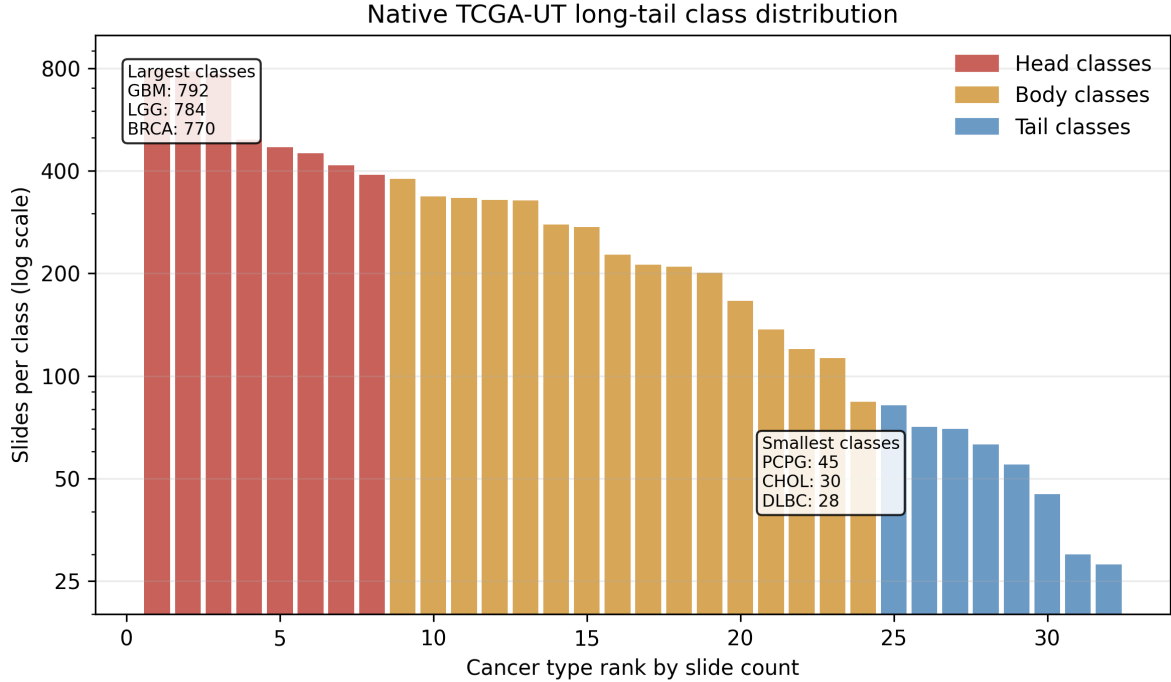


Figure 1: Native TCGA-UT class distribution by cancer-type rank. Bars are grouped into head, body, and tail regions, and the log-scaled y-axis keeps the rare classes visible.

3 Methods

All experiments use frozen Virchow2 features. Virchow and Virchow2 are computational-pathology foundation models trained to encode H&E WSI patches into reusable feature vectors for downstream pathology tasks. The revised representative methods preserve WSI feature bags when the method family depends on slide structure. To fit the available Hydra GPU job memory while keeping all 33 method-seed runs comparable, bag methods use up to 16 evenly spaced feature instances per slide and a bag batch size of 8. Mean-pooled vectors remain only for simple reference baselines such as cross-entropy MLPs, k -nearest neighbors, and nearest-centroid classification. [4, 5]

Table 1 summarizes the method families and the implementation contract used in this experiment. The central rule is that a method is reported as a representative only when its key mechanism is implemented on the available TCGA-UT feature bags. Otherwise, it is described as a baseline or as an excluded image-level extension.

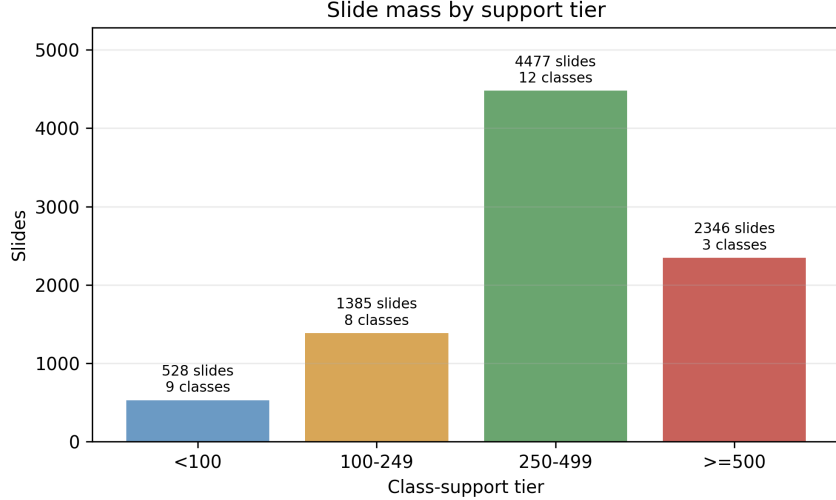


Figure 2: Slide mass by class-support tier. The smallest-support classes are many in label space but contribute little to the total number of training examples.

3.1 Baselines

The reference baselines are six simple methods: unweighted cross-entropy, weighted cross-entropy, focal loss, balanced-sampler cross-entropy, KNN, and nearest-centroid classification. These methods provide context for whether a paper-faithful WSI method improves over simple objective, sampling, or feature-geometry baselines.

3.2 Representative Implementations

The data-level representative is a RankMix-style WSI feature-bag method. It preserves multiple instances per slide, estimates instance importance through attention weights, and adds same-class mixed bag embeddings for examples that are rare or weakly attended. This captures the RankMix premise that WSI augmentation should respect variable bag sizes and class imbalance rather than treating each slide as a fixed image. [6]

The synthetic-generation representative has two layers. First, an image-level stage trains small per-tail-class DCGAN generators on raw TCGA-UT training patch JPEGs and exports generated patch images with a manifest, following the central idea of GAN-based minority image augmentation. Second, the generated patches are re-encoded with Virchow2 and appended to the frozen-feature training bags. To keep this stage inside the Hydra runtime envelope, the image-GAN jobs are parallelized over seed and tail class; each seed produces 4096 generated patches, and the encoder produces 4096 compatible feature tensors that are consumed by the corresponding `feature_gan_mil` run. If the encoded-feature manifest is absent, the classifier records the omission and falls back to the earlier feature-space synthetic condition. This avoids presenting unencoded JPEGs as if they had been used by the frozen-feature WSI classifier. [7, 8]

The algorithm-level representative implements CFAL-style ingredients on bag embeddings: effective-number class weights, class-prototype compactness, and a hard-sample term for embeddings far from their class center. This replaces the earlier batch-logit center penalty with a loss that more directly reflects the affinity and prototype motivation of Center-Focused Affinity Loss. [9]

The hybrid representative follows the MDE-MIL idea through two MIL experts. One expert learns from the original distribution, the other from rebalanced sampling, and the training objective adds a consistency term between their logits. Because no pathology-text encoder artifacts are available in this experiment package, multimodal distillation is treated as a controlled

Table 1: Representative method families used in the revised experiment.

Category	Representative	Implemented mechanism
Data-level	RankMix	Attention-based WSI feature-bag classifier with ranked same-class feature mixing for minority or weakly attended bags. [6]
Synthetic generation	GAN/WGAN augmentation	Tail-class patch-image GAN export plus optional encoded-feature ingestion; image-quality metrics and Virchow2 re-encoding are separate pipeline stages. [7, 8]
Algorithm-level	CFAL	Effective-number class weighting with prototype-oriented affinity and hard-sample compactness penalties on bag embeddings. [9]
Hybrid	MDE-MIL	Two-expert MIL model combining original-distribution and rebalanced objectives with consistency regularization; multimodal distillation is omitted. [10]
Representation	SC-MIL	Attention MIL with supervised contrastive regularization on slide-level bag embeddings. [11]

omission rather than silently approximated.

The representation-level representative follows SC-MIL by combining attention-based MIL with supervised contrastive learning on bag embeddings. This directly targets the feature space, asking whether slides from the same class become more compact and separable under label imbalance. [11]

3.3 Evaluation

Each method is evaluated with seeds 0, 1, and 2. The primary metrics are macro F1, balanced accuracy, accuracy, and classwise recall on the native test split. The result payload also records classwise precision, recall, F1, support, head/body/tail summaries, negative log likelihood, multiclass Brier score, and expected calibration error when probabilities are available. Macro F1 is used as the main ranking metric because it penalizes poor rare-class behavior more directly than accuracy, while support-tier summaries reveal whether gains concentrate in head, body, or tail classes.

4 Results

The full Hydra sweep completed for all 11 methods and all three seeds. The generated missing-results manifest contains an empty missing list, so the aggregate table is based on the complete configured experiment rather than a partial run.

The strongest methods are the simple mean-pooled MLP baselines. Focal loss obtains the highest test macro F1 (0.849) with balanced accuracy 0.846, followed closely by unweighted cross-entropy (macro F1 0.846), balanced-sampler cross-entropy (0.845), and weighted cross-entropy (0.844). Balanced sampling gives the highest balanced accuracy among all methods (0.853), but this does not translate into the highest macro F1.

Among the taxonomy representatives, RankMix-style MIL is the strongest test performer, with macro F1 0.825 and balanced accuracy 0.827. SC-MIL and the encoded image-GAN synthetic condition are close behind at 0.824 and 0.822 macro F1, respectively. The image-GAN bridge completed for all three seeds: it generated 4096 tail-class patches per seed, encoded all 4096 patches per seed with Virchow2, and `feature_gan_mil` loaded all encoded synthetic features without missing files or unknown classes. CFAL reaches 0.818 macro F1, while the

Table 2: Aggregate test performance over three seeds. Methods are sorted by macro F1.

Method	Balanced accuracy	Macro F1	Split
focal	0.846	0.849	test
ce	0.842	0.846	test
balanced_sampler_ce	0.853	0.845	test
weighted_ce	0.844	0.844	test
rankmix_mil	0.827	0.825	test
sc_mil	0.824	0.824	test
feature_gan_mil	0.826	0.822	test
cfal_mil	0.820	0.818	test
mde_mil	0.817	0.804	test
knn	0.768	0.769	test
ncc	0.734	0.686	test

MDE-MIL two-expert adaptation reaches 0.804. The representative implementations are therefore viable and substantially stronger than nonparametric feature-space probes such as KNN (0.769 macro F1) and nearest-centroid classification (0.686), but the completed image-level synthetic path still does not outperform the strongest simple baselines.

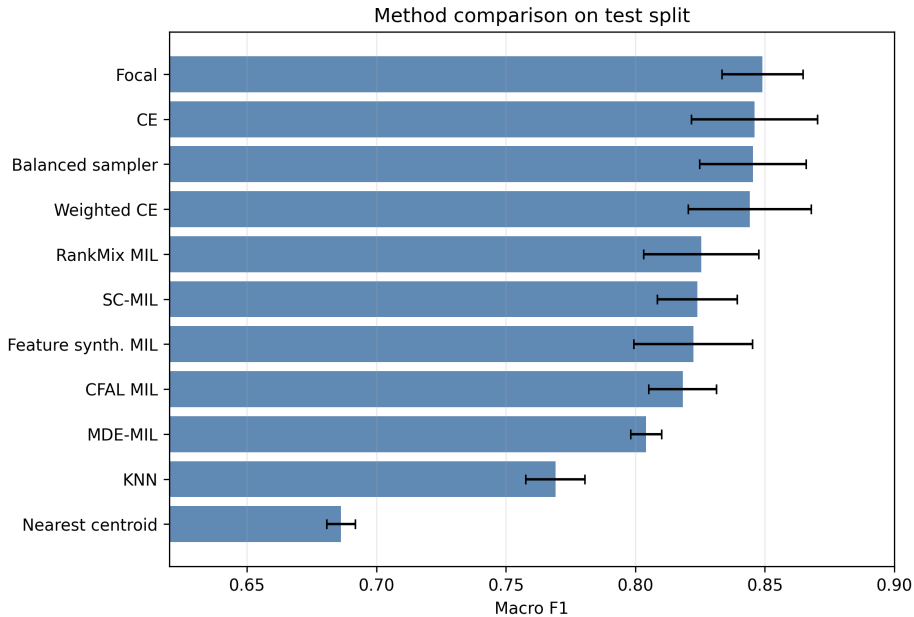


Figure 3: Test macro F1 by method. Error bars summarize seed-to-seed variation across the three configured seeds.

Support-tier summaries explain why balanced accuracy and macro F1 do not always agree. Balanced sampling has the highest tail recall among neural methods (0.787), but its tail F1 is 0.760 because precision falls relative to focal loss. Focal loss has lower tail recall (0.764) but the highest tail F1 among neural methods (0.774), and it also maintains strong head and body F1. MDE-MIL increases tail recall to 0.768, similar to focal loss, but its tail F1 drops to 0.714 and its overall accuracy falls to 0.855, indicating that the two-expert adaptation trades precision and head stability for rare-class sensitivity.

Probability-quality metrics point in the same direction. Focal loss has the lowest mean Brier score among probabilistic neural methods (0.176), whereas the bag representatives range from 0.218 for SC-MIL to 0.263 for MDE-MIL. The encoded image-GAN condition and RankMix-

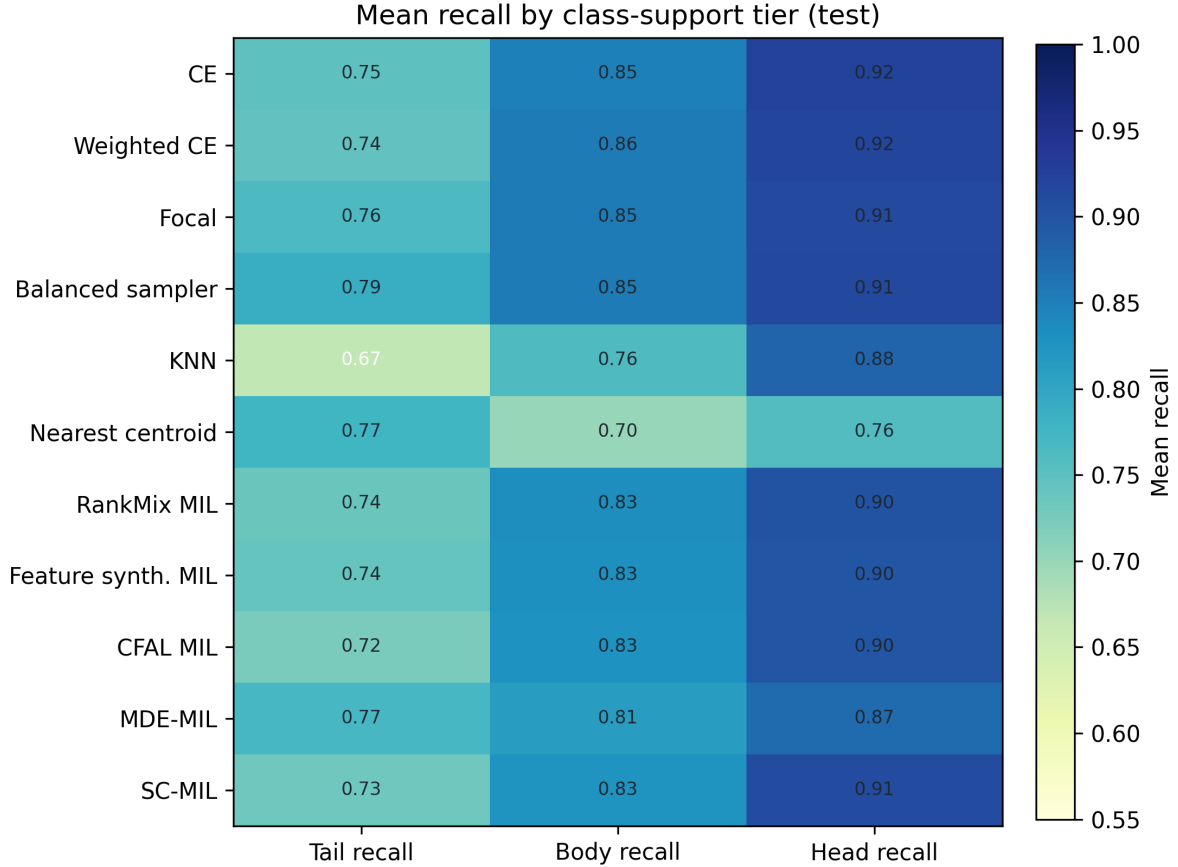


Figure 4: Classwise test recall by method. The figure makes the head–tail variation visible rather than reducing performance to a single aggregate score.

style MIL are therefore not only below the strongest baselines in macro F1, but also less well calibrated under this frozen-feature setup.

5 Discussion

The revised experiment is narrower than a full reproduction of every cited paper, but it is more faithful to the taxonomy than a proxy-only comparison. The study evaluates methods that can be implemented on frozen TCGA-UT WSI feature bags and avoids claiming results for unavailable multimodal components. This matters for interpretation: the new GAN stage generates minority-class patch images and uses them only after a Virchow2 encoding pass, while a two-expert MIL condition can test rebalanced expert learning but cannot evaluate pathology-text distillation without the corresponding encoder.

The main empirical finding is conservative. On frozen Virchow2 features, standard losses and sampling are difficult to beat. This suggests that much of the class-separating signal is already present in the frozen representation and that added MIL structure can be outweighed by adaptation cost, limited bag resolution, or optimization variance. The result should not be read as evidence that RankMix, CFAL, MDE-MIL, or SC-MIL are ineffective in their original settings. It shows that the frozen-feature analogues used here do not outperform simple downstream baselines on TCGA-UT.

The main limitation is therefore not only computational but evidential. Frozen Virchow2 features isolate downstream imbalance mitigation from representation learning, which makes

comparisons controlled, but it also prevents conclusions about end-to-end fine-tuning, patch-level supervision, and image-level generative augmentation. The bag methods also use a 16-instance cap per WSI to keep three-seed Hydra runs inside the available memory envelope. This cap preserves slide-level bags but discards much of the original instance set, so conclusions about attention-based MIL should be treated as conclusions about bounded frozen-feature bags rather than full-resolution WSI bags. Results from this experiment should be read as evidence about downstream WSI feature-bag methods under native TCGA-UT imbalance.

6 Conclusion

This revision turns the TCGA-UT class-imbalance experiment into a taxonomy-grounded comparison with one runnable representative per method family. The complete three-seed sweep shows that focal loss, cross-entropy, balanced sampling, and weighted cross-entropy remain the strongest choices for frozen Virchow2 TCGA-UT features, while the adapted representative bag methods trail by roughly two to five macro-F1 points. The image-GAN bridge confirms that generated tail-class patches can be re-encoded and consumed by the frozen-feature classifier, but this completed synthetic path remains below the strongest simple losses. The study therefore supports a restrained conclusion: for this frozen-feature benchmark, careful baselines are more effective than more elaborate representative adaptations, and multimodal claims require dedicated pipelines rather than feature-space substitutes.

References

- [1] M. Saini and S. Susan, “Tackling class imbalance in computer vision: a contemporary review,” *Artificial Intelligence Review*, vol. 56, no. S1, pp. 1279–1335, 2023.
- [2] Y. Zhang, B. Kang, B. Hooi, and S. Yan, “Deep long-tailed learning: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 9, pp. 10795–10816, 2023.
- [3] C. Mosquera, L. Ferrer, D. H. Milone, D. Luna, and E. Ferrante, “Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance,” *European Radiology*, vol. 34, no. 12, pp. 7895–7903, 2024.
- [4] E. Vorontsov, A. Bozkurt, A. Casson, G. Shaikovski, M. Zelechowski, K. Severson, E. Zimmermann, *et al.*, “A foundation model for clinical-grade computational pathology and rare cancers detection,” *Nature Medicine*, vol. 30, no. 10, pp. 2924–2935, 2024.
- [5] E. Zimmermann, E. Vorontsov, J. Viret, A. Casson, M. Zelechowski, and G. Shaikovski, “Virchow2,” *2024 Shell TechXplorer Digest*, vol. 2024, no. 1, 2024.
- [6] Y.-C. Chen and C.-S. Lu, “Rankmix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 23936–23945, 2023.
- [7] Yuan, “A gan-based data augmentation method for mitigating class imbalance problem in histopathological image classification,” in *2024 IEEE International Conference on Bioinformatics and Biomedicine*, pp. 5327–5334, 2024.
- [8] J. L. Ruiz-Casado, M. A. Molina-Cabello, and M. R. Luque-Baena, “Enhancing histopathological image classification performance through synthetic data generation with generative adversarial networks,” *Sensors*, vol. 24, no. 12, p. 3777, 2024.

- [9] T. Mahbub, A. Obeid, S. Javed, and J. Dias, “Center-focused affinity loss for class imbalance histology image classification,” *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 2, pp. 952–963, 2024.
- [10] X. Ling, Y. Ping, J. Li, J. Peng, and Y. Chen, “Mde-mil,” *Dos mil tres mil*, vol. 27, 2025.
- [11] D. Juyal, S. Shingi, S. A. Javed, H. Padigela, C. Shah, A. Sampat, A. Khosla, J. Abel, and A. Taylor-Weiner, “Sc-mil: Supervised contrastive multiple instance learning for imbalanced classification in pathology,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024.